

The exam contains 13 questions. All questions ask for an evaluation of five statements with (yes/no) or (true/false) answers. A true answer adds +1 points and a false answer adds -1 points to your exam score. Hence, skipping to evaluate a statement is very likely to be more advantageous for you than making a random guess. Your final score is calculated as (Your point sum)/(13*5)*100. The exam duration is four hours.

$P(\cdot)$ denotes a probability measure, probability density function, or probability mass function depending on the context. If not explicitly specified, the difference between these concepts does not affect the answer of the question. $\mathbb{E}_X[f(X)]$ and $\text{Var}_X[f(X)]$ denote the expectation and variance of function f with respect to the distribution of the random variable X , respectively. We use $\mathbb{E}[X]$ as a short-hand for $\mathbb{E}_X[X]$ and $\text{Var}[X]$ for $\text{Var}_X[X]$. All random variables are assumed to have finite expectations. The symbol $\top$ denotes matrix or column vector transpose. While solving the Python implementation questions, assume that the provided code pieces are working properly.

No assumptions may be made additional to those specified in the question text.

Question 1

You are given a population where 1/100 of the population is expected to have cancer. The population is diagnosed by a cancer screening test with a binary outcome. The test is expected to give a negative result in 10% of the cases where the patients actually have cancer. The same test is expected to give a positive result in 3% of the cases where the patients actually do not have cancer. Evaluate the following statements regarding this population and the described cancer test.

Answer 1.1: The probability that a patient with a positive test outcome has cancer is greater than the false positive rate of the test.

Answer 1.2: The probability that a patient will have a positive test outcome is smaller than the probability that a patient will have cancer.

Answer 1.3: The probability that a patient with a negative test outcome does not have cancer is smaller than the probability of the absence of cancer in the whole population.

Answer 1.4: The probability that a patient will be positively tested is smaller than the cancer probability for a patient with a positive test outcome.

Answer 1.5: The joint probability of a patient having cancer and a positive test outcome is smaller than the probability of a patient having cancer conditioned on a positive test outcome.

Question 2

Evaluate the correctness of the following statements regarding two discrete random variables X and Y for all possible joint distributions $P(X, Y)$.

Remark: $\mathbb{E}_Y[f(Y)] = \sum_{y \in \mathcal{Y}} f(y)P(Y = y)$ for a random variable Y defined in domain $\mathcal{Y}$ with a probability mass function $P(Y)$ and a function f that admits inputs from the same domain. The same definition applies also for the random variable X .

Answer 2.1: $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Answer 2.2: $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$

Answer 2.3: $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}_X[X|Y]]$

Answer 2.4: $\text{Var}[X] = \mathbb{E}_Y[\text{Var}_X[X|Y]] + \text{Var}_Y[\mathbb{E}_X[X|Y]]$

Answer 2.5: $\mathbb{E}[X \times Y] = \mathbb{E}[X] \times \mathbb{E}[Y]$

Question 3

Consider the confusion matrix below obtained from a classifier evaluated on medical scans with three possible labels: N (Normal), B (Benign), and M (Malignant). Each of these values corresponds to a class, representing the behavior of a tissue with a suspicion of cancer.

		Prediction			
		N	B	M	
Actual	N	70	20	10	100
	B	10	60	10	80
	M	0	10	10	20
		80	90	30	200

Evaluate the below statements regarding this confusion matrix.

Answer 3.1: The Recall for the Malignant class (M) is 0.50 (or 50%).

Answer 3.2: The Precision for the Benign class (B) is higher than its Recall.

Answer 3.3: The F1-score for the Normal class (N) is exactly 0.75 (or 75%).

Answer 3.4: The model is more likely to provide a false alarm (predicting M when actual is N) than it is to miss a malignant case (predicting either N or B when the actual label is M).

Answer 3.5: The Precision for Class N is higher than the Precision for Class B.

Question 4

Consider a set of n independent and identically distributed (i.i.d.) samples $x_1, x_2, \dots, x_n$ drawn from a Normal distribution where both the mean μ and the variance σ^2 are unknown. Evaluate the below statements regarding the Maximum Likelihood Estimation (MLE) for this case.

Answer 4.1: The MLE for μ is the sample mean $\frac{1}{n} \sum_{i=1}^n x_i$.

Answer 4.2: The MLE for the variance is an unbiased estimator of the true population variance.

Answer 4.3: If the true mean μ were known beforehand, the MLE for the variance would be $\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2$.

Answer 4.4: The log-likelihood function is a linear function of the parameter σ^2 .

Answer 4.5: The maximum value of the likelihood function is achieved at the same pair of (μ, σ^2) values as the maximum of the log-likelihood function.

Question 5

Consider a dataset with 100 samples and two classes (50 Positive, 50 Negative). We are evaluating a potential split on a categorical attribute A that has three possible values: $\{v_1, v_2, v_3\}$. The split results in the following distribution:

- $A = v_1$: 30 samples (20 positive, 10 negative)

- $A = v_2$: 30 samples (10 positive, 20 negative)
- $A = v_3$: 40 samples (20 positive, 20 negative)

Evaluate the below statements regarding this decision tree split.

Answer 5.1: The Gini index of the parent node is exactly 0.5.

Answer 5.2: The Gini index of the child node for $A = v_3$ is higher than the Gini index of the child node for $A = v_1$.

Answer 5.3: The Gini index of the child node for $A = v_1$ is equal to the Gini index of the child node for $A = v_2$.

Answer 5.4: The Information Gain for this split is greater than 0.15 bits.

Answer 5.5: The reduction in Gini index from the parent to the children for this split is greater than 0.04.

Question 6

Consider a Naive Bayes classifier built to predict if a user will buy a certain product ($B = 1$) or not ($B = 0$). The users are characterized by two binary features: Interest category $C \in \{T, F\}$ (where T stands for Tech and F for Fashion) and Discount $D \in \{1, 0\}$ (where 1 means a user has a discount and 0 that the user does not). The training data for 100 users shows:

- 60 users bought the product and 40 users did not.
- Of the 60 users who bought the product, 42 are in interest category T and 18 have a discount.
- Of the 40 users who did not buy the product, 12 are in interest category T and 8 have a discount.

Evaluate the following statements based on the provided data and the Naive Bayes algorithm.

Answer 6.1: Based on the training data, the prior probability of class $B = 1$ is $P(B = 1) = 0.6$.

Answer 6.2: The Naive Bayes assumption implies that the features C and D are marginally independent. In formal terms, it implies $P(C, D) = P(C)P(D)$.

Answer 6.3: The Naive Bayes model will predict that a user from the fashion category ($C = F$) and without a discount ($D = 0$) will buy the product ($B = 1$).

Answer 6.4: The classifier calculates a conditional probability of 0.8 for a user having a discount given they bought the product, denoted as $P(D = 1|B = 1) = 0.8$.

Answer 6.5: The ratio of the posterior probabilities $P(B = 1|C = T, D = 1)/P(B = 0|C = T, D = 1)$ is 14, assuming the prior probabilities are derived directly from the training set frequencies.

Question 7

Consider a decision tree classifier trained on a fixed dataset. We are investigating the effects of varying the hyperparameter `max_depth` (the maximum distance from the root to any leaf) which terminates the splitting process when the maximum depth is reached. Assume the tree is grown using standard recursive partitioning and that the labels are noisy.

Evaluate the below statements regarding the bias-variance trade-off for this model.

Answer 7.1: As the `max_depth` increases toward infinity, the training error generally decreases, but the variance component of the total error typically increases.

Answer 7.2: A decision tree with a `max_depth` of 1 is a higher-variance and lower-bias model compared to decision trees with larger `max_depth`.

Answer 7.3: Increasing the minimum number of samples required to split a node increases model bias.

Answer 7.4: The bias component of the error in a decision tree represents the sensitivity of the model's predictions to small fluctuations in the specific training set used.

Answer 7.5: If a decision tree perfectly classifies every instance in a noisy training set, the total expected error on unseen data is dominated by the variance and the irreducible error (e.g. label noise).

Question 8

```

import torch
import torch.nn as nn

class LogisticRegression(nn.Module):
    def __init__(self, input_dim):
        super(LogisticRegression, self).__init__()
        self.linear = nn.Linear(input_dim, 1)

    def forward(self, x):
        return self.linear(x)

model = LogisticRegression(input_dim=5)
criterion = nn.BCEWithLogitsLoss()
optimizer = torch.optim.SGD(model.parameters(), lr=0.01)

for epoch in range(20):
    for x_batch, y_batch in train_dl:
        logits = model(x_batch.view(-1, 5))
        loss = criterion(logits, y_batch.view(-1, 1))

        loss.backward()
        optimizer.step()
        optimizer.zero_grad()

```

Evaluate the following statements related to the PyTorch source code provided above. **Remark:** The `BCEWithLogitsLoss` function implements the binary cross entropy loss.

Answer 8.1: The `model` instance contains exactly six learnable parameters.

Answer 8.2: It is certain that the code will perform exactly 20 gradient updates.

Answer 8.3: The presented version of the code works without a runtime error for all classification problems regardless of the number of possible class labels as long as the class labels can be predicted from the inputs by a linear mapping.

Answer 8.4: The `loss.backward()` call calculates the gradients for the weights but not for the bias term.

Answer 8.5: The loss function is optimized with respect to the model parameters using the stochastic version of the gradient descent algorithm.

Question 9

Consider the following Python implementation:

```
def learn(self, X, y):
    self.class_prob = numpy.zeros(self.K)
    for k in range(self.K):
        self.class_prob[k] = numpy.sum(y == k) / y.shape[0]
        X_k = X[y == k]
        self.mu[k] = numpy.mean(X_k, axis=0)
        self.Sigma[k] = numpy.cov(X_k.T)

def log_gaussian(self, X, mu, Sigma):
    N, D = X.shape
    log_prob = numpy.zeros(N)
    for i in range(N):
        diff = X[i] - mu
        log_prob[i] = -0.5 * numpy.log(numpy.linalg.det(Sigma)) -
                        0.5 * diff.T @ numpy.
                        linalg.inv(Sigma) @ diff

    return log_prob

def predict(self, X):
    N = X.shape[0]
    log_posterior = numpy.zeros((N, self.K))
    for k in range(self.K):
        log_posterior[:, k] = self.log_gaussian(X, self.mu[k],
                                                self.Sigma[k])
    log_posterior += numpy.log(self.class_prob)
    return numpy.argmax(log_posterior, axis=1)
```

Evaluate the following statements based on the provided code.

Answer 9.1: The implementation calculates a class-specific covariance matrix for each class at an intermediate stage of its computation.

Answer 9.2: The `log_gaussian` function as implemented will correctly compute the logarithm of the probability density for a group of samples where the number of features is 10 but a specific class only has 5 training samples.

Answer 9.3: The `log_gaussian` function as implemented does not return the mathematically exact value of the log-density of a multivariate normal distribution.

Answer 9.4: This implementation performs iterative parameter optimization using the gradient descent algorithm.

Answer 9.5: If the `learn` function were modified to use a shared covariance matrix for all classes, the decision boundaries would be non-linear.

Question 10

Consider a learning algorithm that finds a hypothesis $h \in \mathcal{H}$ by minimizing the empirical error $R_{emp}(h)$ on a training set S of size n , where samples are drawn i.i.d. from a distribution $\mathcal{D}$. Let $R(h)$ be the generalization error. Let $\hat{h}$ be the hypothesis chosen by the empirical risk minimization algorithm on S . Evaluate the following statements.

Answer 10.1: The empirical error $R_{emp}(h)$ is an unbiased estimator of the generalization error $R(h)$ for a fixed hypothesis h , but it is generally a biased estimator for the hypothesis $\hat{h}$ chosen by the empirical risk minimization algorithm.

Answer 10.2: The generalization gap, defined as $|R(h) - R_{emp}(h)|$ for some hypothesis $h \in \mathcal{H}$, is guaranteed to decrease at a linear rate as the number of training samples n increases, regardless of the complexity of a fixed hypothesis space $\mathcal{H}$.

Answer 10.3: For a fixed training set S , if we expand the hypothesis space such that $\mathcal{H} \subseteq \mathcal{H}'$, then it is guaranteed that $R_{emp}(\hat{h}') \leq R_{emp}(\hat{h})$ where $\hat{h}'$ is the hypothesis chosen from $\mathcal{H}'$ by empirical risk minimization.

Answer 10.4: To ensure that $R(\hat{h}) \leq R_{emp}(\hat{h}) + \epsilon$ with confidence $1 - \delta$ for a fixed ϵ and δ , the required number of samples n must scale linearly with the size of the hypothesis space $|\mathcal{H}|$.

Answer 10.5: The inequality $R(\hat{h}) \leq R_{emp}(\hat{h}) + \sqrt{\frac{\ln |\mathcal{H}| + \ln(1/\delta)}{2n}}$, where n is the data set size, holds with probability at least $1 - \delta$ for all finite hypothesis spaces $\mathcal{H}$ but not necessarily for all data distributions.

Question 11

Consider the version of the No-Free-Lunch (NFL) theorem given below:

Let A be any learning algorithm for the task of binary classification with respect to the 0/1 loss over a domain $\mathcal{X}$ and $m < |\mathcal{X}|/2$ be a positive integer representing a training set size. Then, there exists a distribution D over $\mathcal{X} \times \{0, 1\}$ such that:

1. *There exists a function $f : \mathcal{X} \rightarrow \{0, 1\}$ with zero generalization error.*
2. *With probability of at least $1/7$ over the choice of $S \sim D^m$, we have that the generalization error of algorithm A on data set S is at least $1/8$.*

Evaluate the following statements regarding this theorem.

Answer 11.1: If Algorithm A is shown to have smaller generalization error than Algorithm B on a specific dataset, it is expected that its generalization error will be smaller than Algorithm B across all possible data distributions.

Answer 11.2: For any learning algorithm, there exists at least one data distribution for which the algorithm will not have a generalization error smaller than a quantity that cannot be reduced by enlarging the training set size.

Answer 11.3: The theorem demonstrates that inductive bias is a necessary requirement for a learning algorithm to achieve better-than-random generalization on a specific task.

Answer 11.4: The NFL theorem only applies to parametric predictors. It does not explain the behavior of non-parametric models such as the k-nearest-neighbors classifier.

Answer 11.5: NFL implies that for any distribution where a perfect classifier exists, any learning algorithm is guaranteed to converge to zero generalization error as long as the training set size m is large enough to cover at least $1/8$ th of the domain $\mathcal{X}$.

Question 12

Evaluate the following statements about the Probably Approximately Correct (PAC) learning framework.

Answer 12.1: In the PAC framework, a concept is learnable if there exists an algorithm that can find a hypothesis with error less than ϵ with probability at least $1 - \delta$, using a number of samples that grows at a polynomial rate in $1/\epsilon$ for a fixed δ .

Answer 12.2: A larger hypothesis space $\mathcal{H}$ requires fewer training samples to reach the same ϵ error for fixed probability $1 - \delta$.

Answer 12.3: The sample complexity for a finite hypothesis space $\mathcal{H}$ grows at a logarithmic rate in the number of hypotheses available in the hypothesis class ($|\mathcal{H}|$).

Answer 12.4: The confidence parameter δ refers to the maximum rate of incorrect label predictions expected to take place at the generalization time.

Answer 12.5: The Fundamental Theorem of Statistical Learning states that a hypothesis class $\mathcal{H}$ is PAC learnable if and only if its Vapnik-Chervonenkis (VC) dimension is finite.

Question 13

Consider a hypothesis space $\mathcal{H}$ consisting of all possible axis-aligned boxes (rectangular parallelepipeds) in the three-dimensional space of real numbers $\mathbb{R}^3$. A point is classified as $+1$ if it falls inside the box and -1 otherwise. Evaluate the following statements regarding the VC dimension $\text{VC}(\mathcal{H})$ and the shattering of point sets in this space.

Answer 13.1: There exists a set of 6 points in $\mathbb{R}^3$ that can be shattered by $\mathcal{H}$.

Answer 13.2: The VC dimension of $\mathcal{H}$ in $\mathbb{R}^3$ is 4, matching the dimension of the input space plus one.

Answer 13.3: There exists at least one set of four points in $\mathbb{R}^3$ for which every possible labeling can be realized by some axis-aligned box in $\mathcal{H}$.

Answer 13.4: For $n > 10$ where n is the number of points in $\mathbb{R}^3$, the growth function's output value is strictly less than 2^n .

Answer 13.5: There does not exist a set of 5 points in $\mathbb{R}^3$ that can be shattered by $\mathcal{H}$.